Project Proposal: Second-Round Vote Transfer in French Presidential Elections

Data Science Project — Group Proposal

Group composition

Victor Henz, Edgar Gusching, Louis Bettelli, Lucas Magret Delord, Oscar Moulet.

Main question

How do voters of candidates eliminated in the first round of a French presidential election transfer their vote in the second round? (Do they vote for one of the two final candidates or abstain from voting?)

Applied version: given a hypothetical pair of second-round finalists for the 2027 election, how are voters from the other political blocs likely to redistribute their votes department by department, and how much could this shift the national balance?

Data we expect to need

• Historical election results: presidential election results (2002–2022), by polling station, from data.gouv.fr's consolidated elections dataset, aggregated to the department level.

• Geographic reference tables: INSEE's Code Officiel Géographique and the Répertoire Électoral Unique, to map polling stations to communes and departments.

• Socioeconomic indicators: department-level unemployment, income and urban/rural indicators from INSEE and the Observatoire des Territoires, used as explanatory variables for regional variation.

• Original data collection: weekly counts of 2027 candidate mentions in regional French newspapers (via the GDELT news index, or direct scraping of newspapers' search tools if needed), used as a real-time regional media-attention indicator to complement the historical data.

Expected challenges

• Ecological inference: individual vote transfers cannot be observed directly, only inferred from aggregate department-level data, which limits certainty.

• Data heterogeneity: file formats and structures differ across election years and had a documented format change in January 2026, requiring careful harmonization.

• Candidate-to-bloc mapping: classifying candidates into political blocs (far-left to far-right) is a subjective methodological choice that must be documented and justified.

• Statistical noise at department level: smaller departments have noisier vote shares, requiring shrinkage or weighting toward regional averages.

• Media-data coverage risk: GDELT's indexing of French regional dailies is not yet confirmed and will need early validation before committing to it as the sole method.

Deliverable

An interactive web artifact allowing users to select two hypothetical second-round finalists and view the estimated department-by-department vote transfer and its effect on the national balance.